DE-EE0007093
Arizona State University
Atmospheric CO 2 Capture and Membrane Delivery

Milestone 2.1 Basic filter unit can sustain 1 m/s air flow at 20 Pa pressure drop.

A basic filter unit made up of 20 spacers populated with 20 sheets of CO2 sorbent membrane with a vertical spacing of 5 mm (see insert) were assembled and placed in the test chamber of an Omega WT 4401D-D wind tunnel.

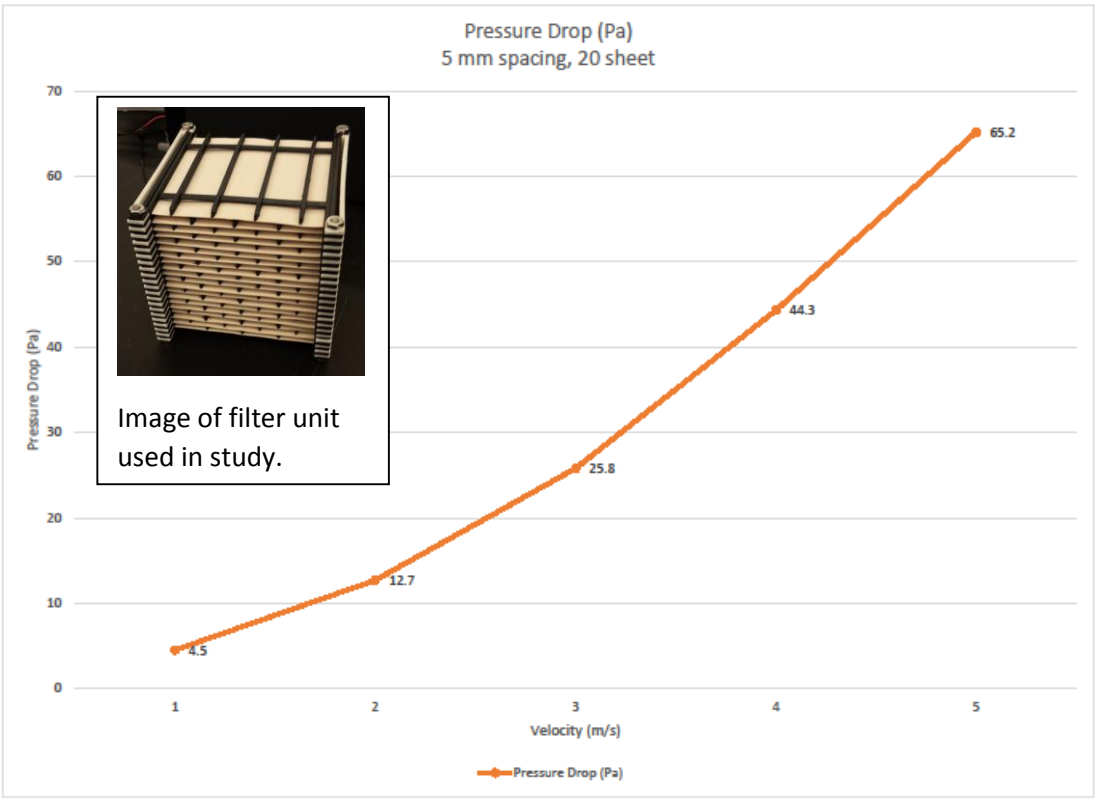


Image of filter unit used in study.

It is illustrated on the graph that a velocity of 1 m/s sees a pressure drop of 4.5 Pa. The velocity required to create a pressure drop of 20 Pa is approximately 2.5 m/s, more than double the test velocity. We therefore demonstrate that Milestone 2.1, the ability of a filter module to sustain a wind velocity of 1 m/s with a pressure drop of less than 20 Pa has been successfully met.